Bulacan State University
City of Malolos, Bulacan

Educational Development Office

BACHELOR OF SCIENCE IN CIVIL ENGINEERING
Specialized Track: Structural Engineering
Revised 2024
CMO 92 series of 2017

PROGRAM OF STUDY

FIRST YEAR

First Semester

Code	Course Description	Lab. Units	No. of Lab. Hours	Lect. Units	No. of Lect. Hours	Pre-Req	Co-Req
MMW 101	Mathematics in Modern World	0	0	3	3		
UTS 101	Understanding the Self	0	0	3	3		
PAL 101	Panitikan at Lipunan	0	0	3	3		
PCM 101	Purposive Communication	0	0	3	3		
ETH 101	Ethics	0	0	3	3		
COE 100	Mathematics for Engineers	0	0	3	3		
COE 101	Introduction to Engineering	0	0	1	1		
COE 102	Calculus 1 (Differential Calculus)	0	0	3	3		
COE 103	Chemistry for Engineers - Lecture	0	0	3	3		
COE 103L	Chemistry for Engineers - Laboratory	1	3	0	0		COE 103
PE 10	PATHFit 1	0	0	2	2		
NSTP 10	National Service Training Program 1	0	0	3	3		
	Total	**1**	**3**	**30**	**30**		

Second Semester

Code	Course Description	Lab. Units	No. of Lab. Hours	Lect. Units	No. of Lect. Hours	Pre-Req	Co-Req
RPH 101	Readings in Philippine History	0	0	3	3		
TCW 101	The Contemporary World	0	0	3	3		
AAP 101	Arts Appreciation	0	0	3	3		
COE 104	Calculus 2 (Integral Calculus)	0	0	3	3	COE 102	
COE 105	Physics for Engineers - Lecture (Calculus Based)	0	0	3	3	COE 102	COE 104; COE 105L
COE 105L	Physics for Engineers - Laboratory (Calculus Based)	1	3	0	0		COE 105
CIV 101	Mathematics for Civil Engineers	0	0	3	3		
CIV 102	Civil Engineering Orientation	0	0	2	2		
CIV 103D	Engineering Drawing and Plans	1	3	0	0		
CIV 104	Computer Fundamentals and Programming	2	6	0	0	COE 102	
PE 11	PATHFit 2	0	0	2	2	PE 10	
NSTP 11	National Service Training Program 2	0	0	3	3	NSTP 10	
	Total	**4**	**12**	**25**	**25**		

SECOND YEAR

CONTENTS NOTED
DR. MARIA TERESA S. SALTA
Education Supervisor II
Date: 7-10-2024

First Semester							
Code	**Course Description**	**Lab. Units**	**No. of Lab. Hours**	**Lect. Units**	**No. of Lect. Hours**	**Pre-Req**	**Co-Req**
COE 201	Differential Equations	0	0	3	3	COE 104	
COE 202	Engineering Data Analysis	0	0	3	3	COE 104	
CIV 201	Statics of Rigid Bodies	0	0	3	3	CIV 101; COE 104; COE 105; COE 105L	
CIV 202	Fundamentals of Surveying 1 (Lecture)	0	0	3	3	CIV 104D	CIV 202F
CIV 202F	Fundamentals of Surveying 1 (Field)	1	3	0	0		CIV 202
CIV 203	Geology for Civil Engineers	0	0	2	2	COE 103; COE 103L	
CIV 204	Building Systems Design (Lecture)	0	0	2	2	CIV 103D	CIV 204D
CIV 204D	Building Systems Design (Design)	1	3	0	0		CIV 204
CIV 205L	Computer-Aided Drafting (Laboratory)	1	3	0	0	CIV 103D; CIV 104	
SSP 101d	The Entrepreneurial Mind	0	0	3	3		
PE 12	PATHFit 3	0	0	2	2	PE 11	
	Total	**3**	**9**	**21**	**21**		
Second Semester							
Code	**Course Description**	**Lab. Units**	**No. of Lab. Hours**	**Lect. Units**	**No. of Lect. Hours**	**Pre-Req**	**Co-Req**
CIV 206	Dynamics of Rigid Bodies	0	0	2	2	CIV 201	CIV 207
CIV 207	Mechanics of Deformable Bodies	0	0	4	4	CIV 201	CIV 206
CIV 208	Numerical Solutions to CE Problems (Lecture)	0	0	2	2	COE 201	CIV 208L
CIV 208L	Numerical Solutions to CE Problems (Laboratory)	1	3	0	0		CIV 208
CIV 209F	Fundamentals of Surveying 2 (Field)	1	3	0	0	CIV 202; CIV 202F	
CIV 210	Engineering Economics	0	0	3	3	COE 202	
CIV 211	Engineering Utilities 1	0	0	3	3	COE 105; COE 105L; CIV 204; CIV 204D	
CIV 212	Engineering Utilities 2	0	0	3	3	COE 105; COE	

CONTENTS NOTED

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						105L; CIV 204; CIV 204D	
AE 101	Academic English	0	0	3	3		
STS 101	Science, Technology and Society	0	0	3	3		
PE 13	PATHFit 4	0	0	2	2	PE 12	
	Total	**2**	**6**	**25**	**25**		

THIRD YEAR

First Semester

Code	Course Description	Lab. Units	No. of Lab. Hours	Lect. Units	No. of Lect. Hours	Pre-Req	Co-Req
CIV 301	Structural Theory 1 (Lecture)	0	0	2	2	CIV 207	CIV 301D
CIV 301D	Structural Theory 1 (Design)	1	3	0	0		CIV 301
CIV 302	Soil Mechanics (Lecture)	0	0	3	3	CIV 203; CIV 207	CIV 302L
CIV 302L	Soil Mechanics (Laboratory)	1	3	0	0		CIV 302
CIV 303	Hydraulics 1	0	0	3	3	CIV 206; CIV 207	
CIV 304	Construction Material and Testing (Lecture)	0	0	2	2	CIV 206; CIV 207	CIV 304L
CIV 304L	Construction Material and Testing (Laboratory)	1	3	0	0		CIV 304
CIV 305	Highway and Railroad Engineering	0	0	3	3	CIV 209F; CIV 206	
CIV 306	Engineering Management	0	0	2	2	COE 202; CIV 210	
CIV 307	Environmental Engineering	0	0	2	2	CIV 203	
CIV 308	Quantity Surveying (Lecture)	0	0	1	1	CIV 211; CIV 212	CIV 308L
CIV 308L	Quantity Surveying (Laboratory)	1	3	0	0		CIV 308
	Total	**4**	**12**	**18**	**18**		

Second Semester

Code	Course Description	Lab. Units	No. of Lab. Hours	Lect. Units	No. of Lect. Hours	Pre-Req	Co-Req
CIV 309	Structural Theory 2 (Lecture)	0	0	2	2	CIV 301; CIV 301D	CIV 309D
CIV 309D	Structural Theory 2 (Design)	1	3	0	0		CIV 309
CIV 310	Hydraulics 2 (Lecture)	0	0	2	2	CIV 303	CIV 310L

CONTENTS NOTED

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CIV 310L	Hydraulics 2 (Laboratory)	1	3	0	0		CIV 310
CIV 311	Construction Methods and Project Management	0	0	3	3	CIV 306; CIV 308; CIV 308L	
CIV 312	Foundation Engineering (Lecture)	0	0	2	2	CIV 302; CIV 302L	CIV 312L
CIV 312L	Foundation Engineering (Laboratory)	1	3	0	0		CIV 312
CIV 313	Technopreneurship 101	0	0	3	3	CIV 306; SSP 101d	
CIV 314	Principles of Transportation Engineering	0	0	3	3	CIV 305	
CIV 315	Hydrology	0	0	3	3	COE 202; CIV 302; CIV 302L; CIV 303	
CIV 316	Earthquake Engineering (Design)	1	3	0	0	CIV 203; CIV 301; CIV 301D	
	Total	**4**	**12**	**18**	**18**		

Summer Term

Code	Course Description	Lab. Units	No. of Lab. Hours	Lect. Units	No. of Lect. Hours	Pre-Req	Co-Req
OJT 400*	SUMMER OJT On-The-Job Training (250 Hours)	2	250	1	1	CIV 302; CIV 302L; CIV 304; CIV 304L; CIV 309; CIV 309D; CIV 310; CIV 310L; CIV 311; CIV 314	
	Total	**2**	**250**	**1**	**1**		

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DR. MARIA TERESA S. SALTA
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Date: 7-19-2024

FOURTH YEAR

First Semester

Code	Course Description	Lab. Units	No. of Lab. Hours	Lect. Units	No. of Lect. Hours	Pre-Req	Co-Req

CIV 401	Structural Steel and Timber Design (Lecture)	0	0	3	3	CIV 309; CIV 309D	CIV 401D
CIV 401D	Structural Steel and Timber Design (Design)	1	3	0	0		CIV 401
CIV 402	Reinforced Concrete Design (Lecture)	0	0	3	3	CIV 309; CIV 309D	CIV 402D
CIV 402D	Reinforced Concrete Design (Design)	1	3	0	0		CIV 402
CIV 403	Civil Engineering Project Part 1 (Lecture)	0	0	1	1	CIV 302; CIV 302L; CIV 304; CIV 304L; CIV 307; CIV 309; CIV 309D; CIV 310; CIV 310L; CIV 311; CIV 313; CIV 314; PCM 101	CIV 403D
CIV 403D	Civil Engineering Project Part 1 (Design)	1	3	0	0	CIV 302; CIV 302L; CIV 304; CIV 304L; CIV 307; CIV 309; CIV 309D; CIV 310; CIV 310L; CIV 311; CIV 313; CIV 314; PCM 101	CIV 403
CIV 404	Water Resources Engineering	0	0	2	2	CIV 310; CIV	

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						310L; CIV 315	
CIV 405	Correlation 1	1	3	0	0	CIV 101; CIV 302; CIV 302L; CIV 308; CIV 308L; CIV 309; CIV 309D; CIV 310; CIV 310L; CIV 314	
CIV 410-STE	Matrix Analysis of Structures (SPECIALIZATION)	0	0	3	3	CIV 208; CIV 208L; CIV 309; CIV 309D	
CIV 411-STE	Computer Software in Structural Analysis (SPECIALIZATION)	0	0	3	3	CIV 208; CIV 208L; CIV 309; CIV 309D	CIV 401; CIV 401D; CIV 402; CIV 402D
	Total	**4**	**12**	**15**	**15**		

Second Semester

Code	Course Description	Lab. Units	No. of Lab. Hours	Lect. Units	No. of Lect. Hours	Pre-Req	Co-Req
CIV 406	Civil Engineering Project Part 2 (Lecture)	0	0	1	1	CIV 403; CIV 403D	CIV 406D
CIV 406D	Civil Engineering Project Part 2 (Design)	1	3	0	0	CIV 403; CIV 403D	CIV 406
CIV 407F	Seminars and Plant Visits	1	3	0	0	CIV 313	
CIV 408	CE Laws, Ethics and Contracts	0	0	2	2	ETH 101; CIV 311	
CIV 409	Correlation 2	1	3	0	0	CIV 312; CIV 312L; CIV 401; CIV 401D;	CIV 406; CIV 406D; CIV 407F; CIV 408;

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Date: 7-14-2021

						CIV 402; CIV 402D; CIV 405; CIV 410-STE; CIV 411-STE	CIV 412-STE; CIV 413-STE; CIV 414-STE
CIV 412-STE	Foundation and Retaining Wall Design (SPECIALIZATION)	0	0	3	3	CIV 312; CIV 312L; CIV 402; CIV 402L	
CIV 413-STE	Special Topics (SPECIALIZATION)	0	0	3	3	CIV 309; CIV 309D; CIV 401; CIV 401D; CIV 402; CIV 402D	
CIV 414-STE	Prestressed Concrete Design (SPECIALIZATION)	0	0	3	3	CIV 402; CIV 402D	
RLW 101	Life and Works of Rizal	0	0	3	3		
	Total	**3**	**9**	**15**	**15**		

Summary: Legal Basis: (**CMO 92 series 2017**)

Courses	CHED Requirements (no. of units)	BulSU 2024 Revised Curriculum (no. of units)
I. TECHNIAL COURSES		
A. Mathematics	15	15
B. Natural/Physical Sciences	10	10
C. Basic Engineering Sciences	23	23
D. Allied Courses	6	6
E. Professional Courses-Common	49	54
F. Professional Courses - Specialized	15	15
G. On-the-Job Training (minimum of 240 hours)	3	3
H. Additional Courses	0	19
II. NON-TECHNIAL COURSES		
General Education	24	24
GEC Elective/Mandated Courses	12	12
PE (PATHFIT)	8	8
NSTP	6	6
TOTAL	171	195

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Education Supervisor II
Date: 7-10-2024